Visualizing Global Lead Pollution Policy

A Comparative Study of Lead Paint Regulation

By: Mohammad A. Hijazi

MS Data Science at LAU – School of Arts and Sciences

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Abstract

Lead pollution remains a significant global public-health concern, especially because lead exposure can affect multiple body systems and is particularly harmful to children and more vulnerable individuals. The World Health Organization (WHO) states that there is no known safe level of lead exposure, and "Our World in Data" (OWID) describes lead pollution as a widespread but under-recognized global burden. This project develops an interactive visualization dashboard to analyze global disparities in legally binding lead paint regulation and to connect regulation status with lead-attributable health-burden indicators, specifically DALY rate and death rate. The dashboard uses country-level lead paint regulation data from OWID and health-burden data derived from IHME Global Burden of Disease 2023. Besides, the project follows a research-driven visualization study involving problem formulation, data abstraction, encoding justification, validation reasoning, critical evaluation, and professional communication.

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1. Introduction and Problem Formulation

1.1 Topic Motivation

Lead pollution is a global environmental health issue that continues to affect populations despite decades of progress in eliminating some major sources of exposure. OWID notes that lead pollution receives limited attention despite contributing substantially to global disease burden and affecting children's brain development, cognition, and long-term wellbeing. The WHO similarly emphasizes that lead exposure can affect multiple body systems, is especially harmful to young children, and has no known safe exposure level.

One preventable source of such exposure is **lead paint**. OWID states that it's a main contributor to harmful lead exposure, and the lead paint regulation dataset explicitly tracks which countries have legally binding controls on lead paint. WHO explains that legally binding lead paint controls can help prevent childhood and occupational exposure to lead in new paints and reduce future legacy exposure from such paints.

This project is motivated by the fact that the lead paint regulation remains uneven globally. WHO reports that, as of 16 January 2024, less than 50% of countries had officially confirmed legally binding controls on the production, import, sale, and use of lead paints. This makes the topic suitable for such a data visualization project because the main problem is not only statistical but also communicative: policy gaps, regional disparities, and health-burden patterns need to be made visible to support interpretation and decision-making.

1.2 Analytical Problem

The central analytical problem is: How can global lead paint regulation status be visually analyzed and communicated, and how does regulation status appear alongside lead-attributable DALY and death-rate patterns across countries and regions?

This problem is complex because the raw data is country-level, categorical, geographic, and health-related. A simple table can list regulation status, but it does not easily reveal spatial clusters, regional disparities, high-burden countries, or policy-health pattern relationships. However, this project transforms country-level regulatory and health-burden data into an interactive visual dashboard.

In other words, this project aims in answering more-practically the following questions: which countries have legally binding lead paint controls, where major geographic gaps exist, and whether combining regulation data with lead-related health indicators can reveal broader policy-health patterns.

2. Task Abstraction

Here, the listed-previously domain problem is transformed into the following visualization tasks:

Table 1: A table showing the project's task abstraction

Domain Question	Abstract Visualization Task	Dashboard Component
Which countries are regulated or not regulated?	Lookup and identify categorical status	Choropleth world map
Where are geographic policy gaps?	Spatial pattern detection	Choropleth world map
How many countries are regulated?	Summarize and compare categorical counts	Regulation count bar chart with KPI cards
Which regions have higher regulation coverage?	Compare proportions across groups	Horizontal 100% stacked regional bar chart
How do DALY and death rates relate?	Explore relationship between two quantitative variables	Scatterplot with trend lines
Which countries have the highest lead burden?	Rank and compare quantitative values	Ranked lollipop charts

How do regions compare across multiple indicators?	Multivariate regional summarization	Regional heatmap

These tasks compose the core logic of this project, which will lead to answering to the important questions of this topic relevantly.

3. Data Description, Preparation, and Abstraction

3.1 Data Sources

The project integrates two main data sources:

1. **Lead paint regulation data** from OWID's grapher dataset, which records whether each country has legally binding controls on lead paint, using categories such as "Yes" and "No."
2. **Lead-attributable health-burden data** from the IHME Global Burden of Disease 2023 data, specifically death rate and DALY rate attributed to lead exposure. IHME's GBD 2023 resources provide global estimates for countries and territories, including lead exposure estimates across years, locations, age groups, and sex categories.

It's planned to use the OWID lead paint regulation dataset as the core source and to optionally integrate additional lead-related indicators where compatible country-level data were available, which is the second listed dataset in the-project's case. Besides, it's planned to merge the listed regulation status with both DALY and death-rate indicators.

3.2 Data Structure

The processed dataset is produced by the Python file "prepare_dataset.py". This dataset contains one row per country and includes the following main fields:

- country: standardized country name.
- iso_code: ISO-3 country code.
- Year: year of regulation status.

- regulation: original binary category, “Yes” or “No”.
- regulation_label: display category, “Regulated” or “Not regulated”.
- region: derived continent-level region.
- death_rate: lead-attributable death rate.
- daly_rate: lead-attributable DALY rate.
- ihme_country: original IHME country name, retained for traceability purposes.

This preparation script filters IHME data to 2023, all ages, both sexes, and rate metrics, then extracts “Deaths” and “DALYs” from the long-format IHME dataset. It also standardizes country names, fixes selected ISO-code issues when necessary, derives region from ISO-3 codes, and saves the final dataset as “data/processed/lead_policy_merged.csv”.

3.3 Data Preparation Steps

The data preparation involved the following steps:

1. **Filtering health-burden data:** IHME GBD data was filtered to 2023, all ages, both sexes, and rate-based metrics. This creates a consistent cross-country comparison basis.
2. **Extracting deaths and DALYs:** Because IHME stores deaths and DALYs as rows under “measure_name”, the script extracted death rate and DALY rate separately and then merged them into a wide country-level table.
3. **Country-name harmonization:** The script used a crosswalk to align IHME dataset names such as “Lao People's Democratic Republic”, “Russian Federation”, “United States of America”, and “Viét Nam” with regulation dataset names such as “Laos”, “Russia”, “United States”, and “Vietnam”.
4. **ISO-code correction:** The script handled country-name fixes such as “Timor” to “Timor-Leste” and assigned missing ISO codes such as TLS for Timor-Leste.

5. **Region derivation:** The dashboard required regional comparison, but the original data did not include a "region" column. This was derived from ISO-3 country codes using "pycountry" and "pycountry_convert" Python libraries, with manual fixes for special cases such as listed-above Timor-Leste.
6. **Display-label creation:** The original "Yes/No" variable was mapped to "Regulated/Not regulated" to improve dashboard readability.

3.4 Data Abstraction

From a visualization perspective, the variables are abstracted as follows:

Table 2: A table showing the project's data abstraction.

Variable	**Data Type**	**Visualization Role**
Country	Nominal	Geographic and categorical entity
ISO code	Nominal identifier	Spatial mapping
Region	Derived categorical variable	Grouping and comparison
Regulation status	Binary categorical	Color hue and grouping
DALY rate	Quantitative	Position, ranking, heatmap value
Death rate	Quantitative	Position, ranking, heatmap value

This abstraction is appropriate because the project combines spatial, categorical, and quantitative comparisons.

4. Dashboard Overview and Implementation

The dashboard was implemented in Python using both "Dash" and "Plotly" packages. The main app (i.e. "app.py") loads the processed dataset and calls "enhanced_dashboard_layout(df)" from "dashboard_visual_enhancements.py", separating data preparation from visualization layout and chart functions.

The dashboard includes:

- KPI cards summarizing total countries, regulated countries, unregulated countries, and average DALY and death rates.
- A global choropleth map showing country-level regulation status.
- A bar chart comparing the number of regulated and not-regulated countries.
- A horizontal 100% stacked bar chart showing regulation share by region.
- A scatterplot showing the relationship between DALY and death rates.
- Two ranked lollipop charts showing top countries by DALY and death rates.
- A regional heatmap summarizing regulation percentage, average DALY rate, and average death rate.

5. Encoding Design: Marks and Channels

What's done as follow is identifying marks, visual channels, perceptual justification, and alternatives rejected for each major visualization.

5.1 KPI Cards

Marks: Text marks.
Channels: Position, size, typography, grouping.
Purpose: Provide a high-level summary before detailed exploration.

The KPI cards use large numbers and short labels to communicate global totals quickly. This supports overview-first interpretation before the user examines detailed charts.

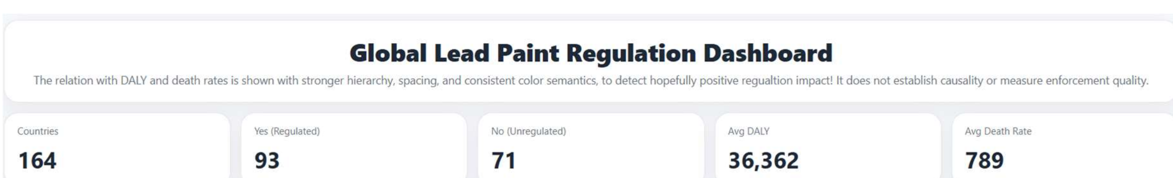

Figure 1: KPI cards showing country count, regulated countries, unregulated countries, average DALY rate, and average death rate.

5.2 Choropleth World Map

Marks: Country polygon areas.
Channels: Geographic position and color hue.
Purpose: Show spatial distribution of lead paint regulation.

Color hue is appropriate because regulation status is binary categorical. The dashboard uses blue for “Regulated” and red for “Not regulated,” maintaining consistent semantics across all charts. The map is appropriate because the analytical task is geographic pattern detection.

Alternative rejected: A table was rejected as the primary view because it supports lookup but not spatial pattern recognition. A bar chart alone was also insufficient because it cannot show geographic clustering.

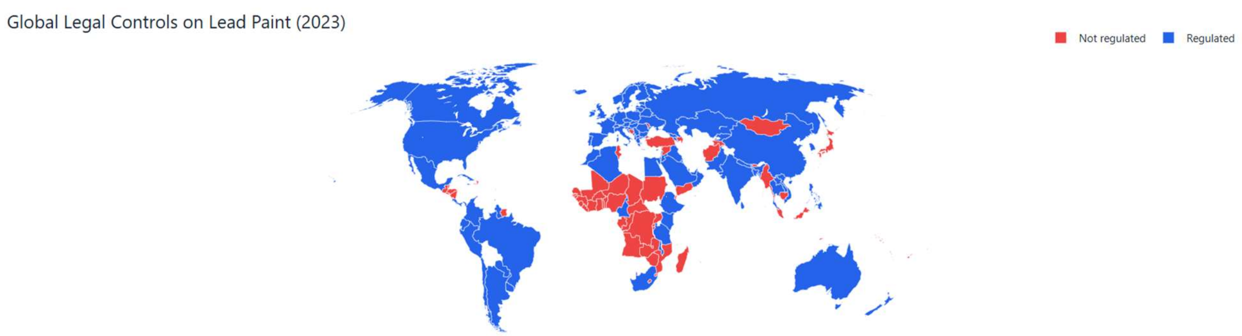

Figure 2: Choropleth map of legally binding lead paint controls by country in 2023.

5.3 Country Count Bar Chart

Marks: Bars.
Channels: Length, position, color hue, text labels.
Purpose: Compare total number of regulated and not-regulated countries.

Length on a common baseline is highly effective for comparing quantities. The chart complements the map by answering “how many?” rather than “where?”

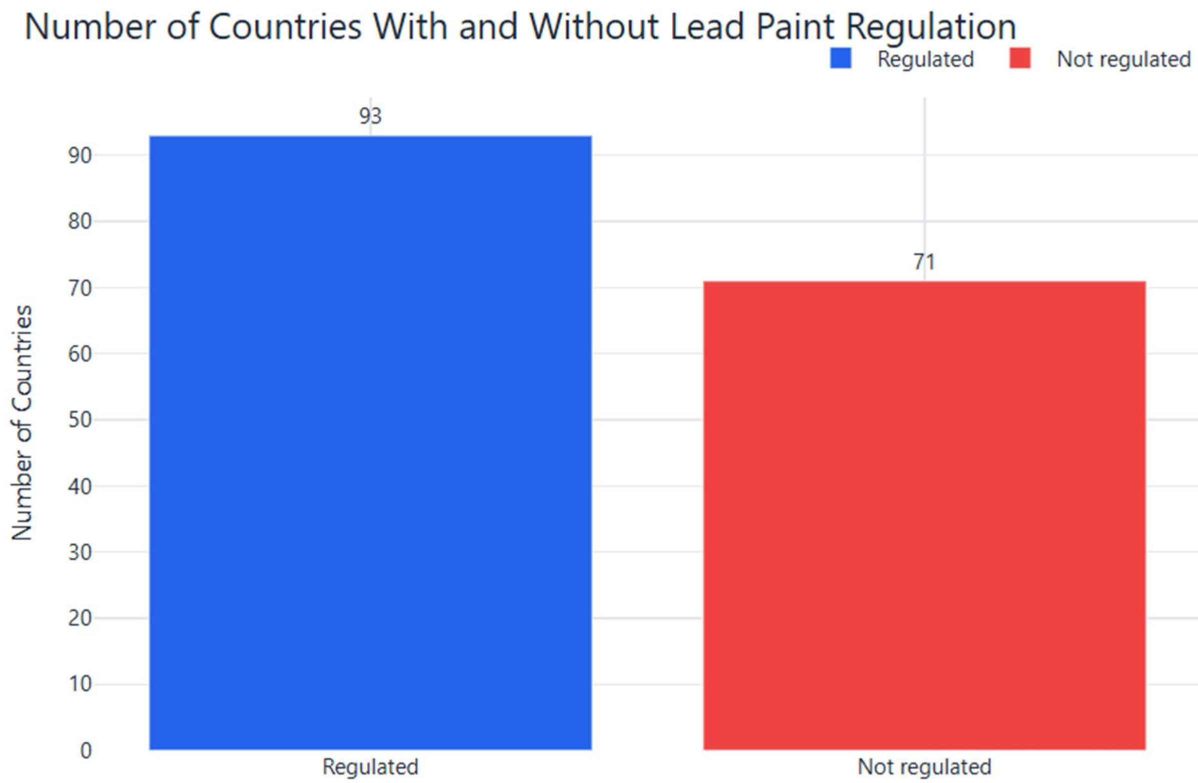

Figure 3: Number of countries with and without lead paint regulation.

5.4 Regional 100% Stacked Bar Chart

Marks: Horizontal stacked bars.
Channels: Length, position, color hue, text labels.
Purpose: Compare the proportion of regulated and not-regulated countries across regions.

A horizontal layout was used to avoid rotated region label-names and improve readability. Regions are sorted by the percentage of regulated countries, allowing the viewer to identify regional policy disparities quickly.

Alternative rejected: A vertical stacked bar chart was tested precisely, resulting in rotated labels to avoid overlapping of the words used in labels (because some label-names are long), but this reduced readability. The horizontal version better supports regional comparison.

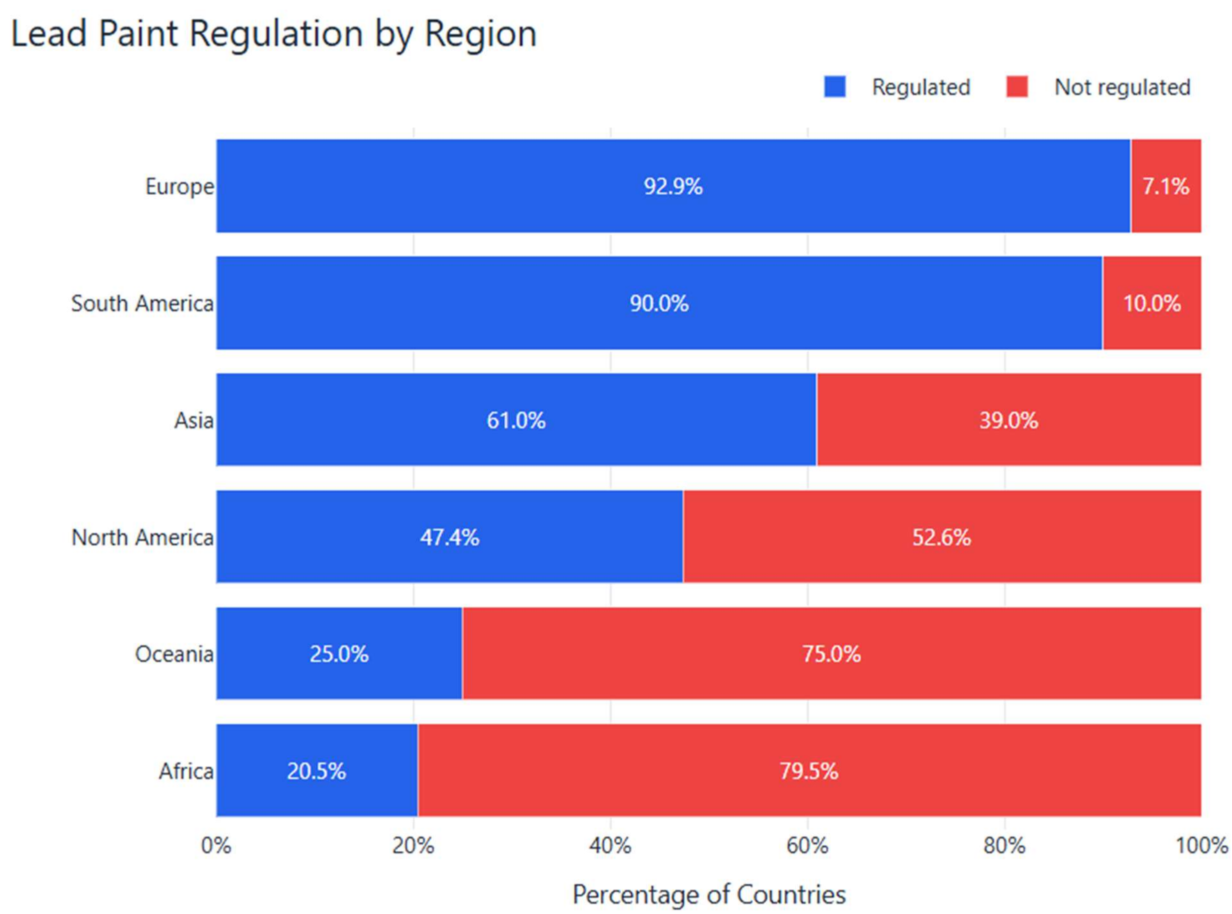

Figure 4: Regional lead paint regulation share, sorted by percentage regulated.

5.5 Scatterplot of DALY Rate vs Death Rate

Marks: Points and trend lines.
Channels: X-position, Y-position, color hue, line slope.
Purpose: Explore the relationship between lead-attributable DALY rate and death rate.

The scatterplot uses position on two quantitative axes, which is effective for relationship analysis. Color hue indicates regulation status, allowing the viewer to compare regulated and not-regulated countries within the health-burden relationship.

The dashboard displays a positive correlation between DALY rate and death rate, showing that countries with higher DALY burden tend to also have higher death-rate burden. However, this should be interpreted <u>as association only</u>, not causality.

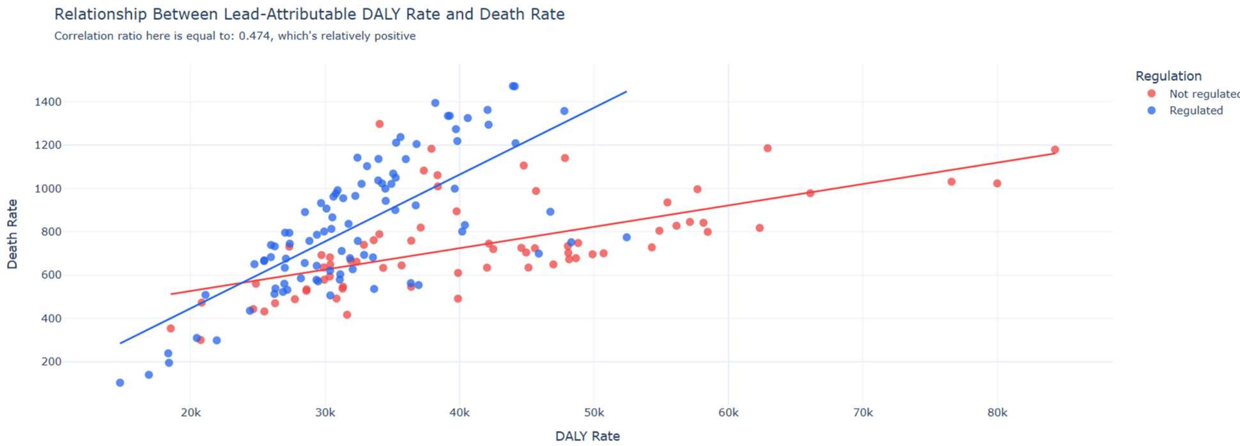

Figure 5: Relationship between lead-attributable DALY rate and death rate, colored by regulation status.

5.6 Ranked Lollipop Charts

Marks: Lines and points.
Channels: Horizontal position, vertical position, color hue, direct value labels.
Purpose: Rank countries by lead-attributable burden.

The lollipop charts show the top countries by DALY rate and death rate. Horizontal position encodes burden magnitude, vertical position encodes country rank, and color hue encodes regulation status. Direct value labels reduce dependence on tooltips and improve readability.

Alternative rejected: A full bar chart was possible, but the lollipop format reduces visual weight and emphasizes ranked points more clearly.

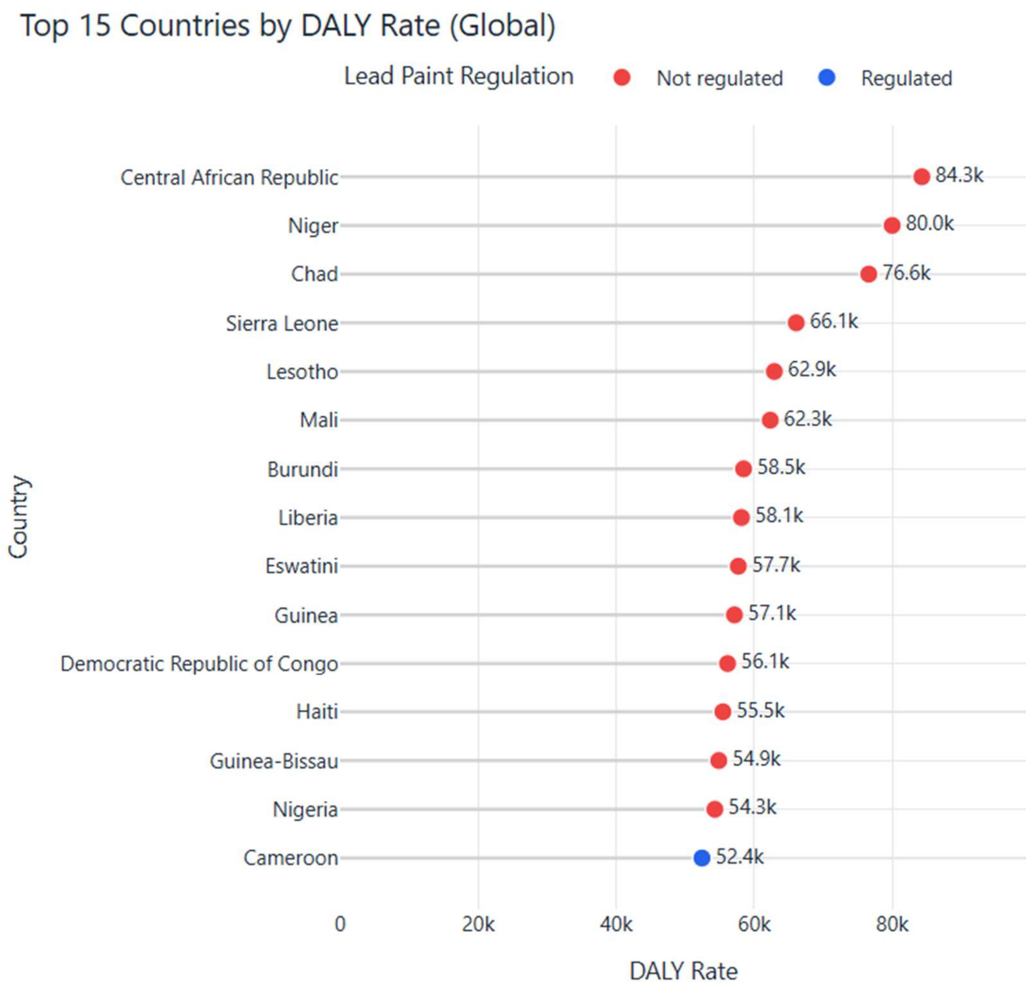

Figure 6: Ranked lollipop chart of top countries by DALY rate.

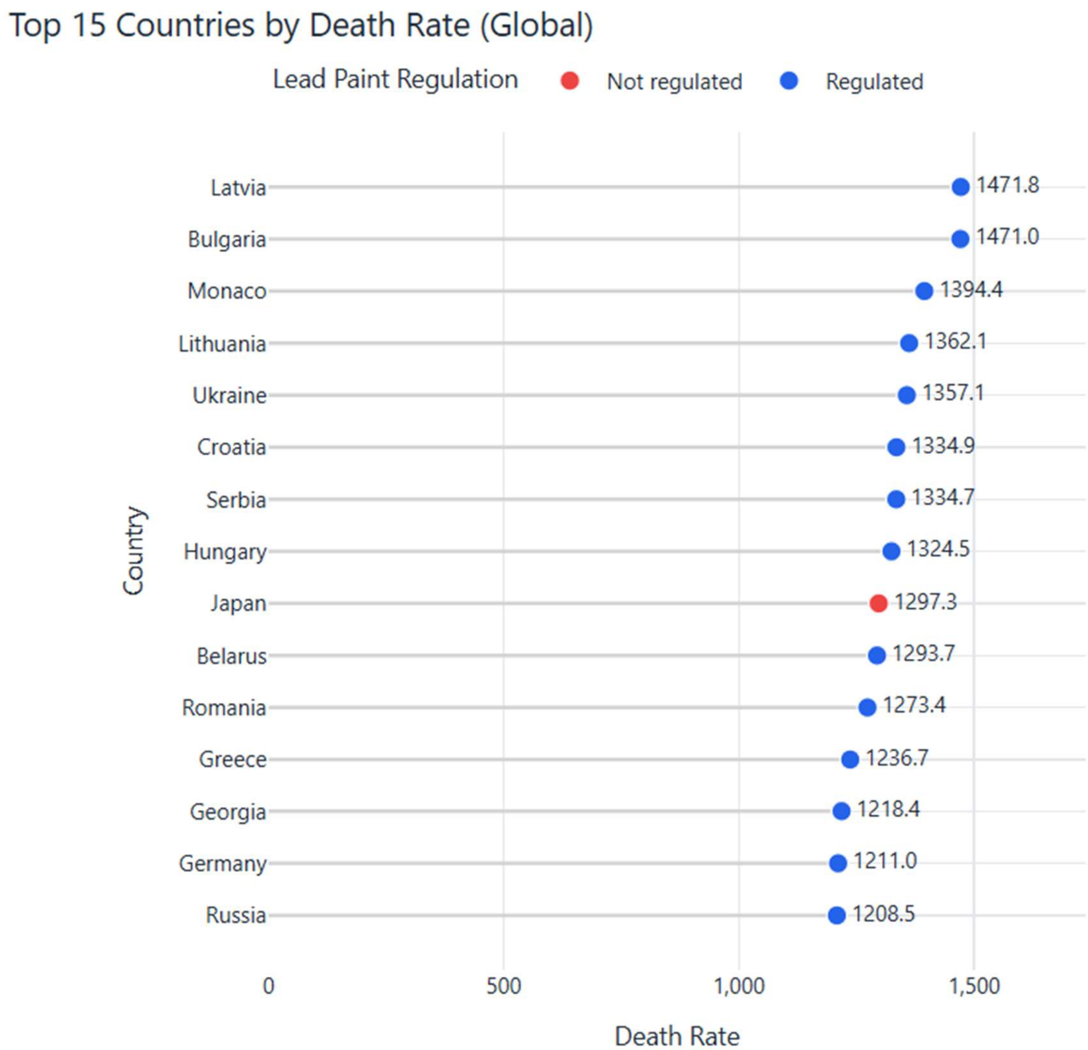

Figure 7: Ranked lollipop chart of top countries by death rate.

5.7 Regional Heatmap

Marks: Rectangular cells.
Channels: Color intensity (discretely), position, text labels.
Purpose: Summarize multiple indicators by region.

The heatmap uses region on one axis and indicator on the other. Because the indicators have different units, the dashboard normalizes values for color intensity while displaying actual values as text. This avoids misleading comparisons across incomparable units. Besides, the color scale for these values is the one dividing into bins, to avoid possible vision-related brain overloading due to a very large number of colors to take into consideration when viewing this heatmap.

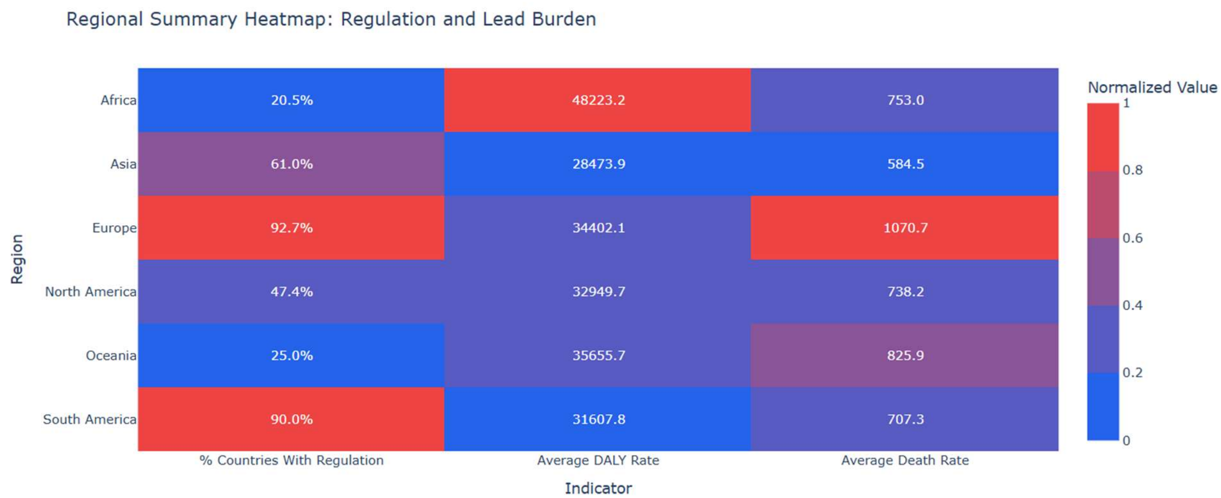

Figure 8: Regional heatmap summarizing regulation share, average DALY rate, and average death rate.

6. Rules of Thumb and Chart Critique

The dashboard follows several visualization rules of thumb: clarity, cognitive efficiency, chart selection, and critical comparison of alternatives.

First, categorical regulation status is consistently encoded using color hue: blue represents regulated countries and red represents not-regulated countries. This avoids color-semantic confusion across charts, and also takes into consideration all categories of viewers including ones with color blindness. Second, quantitative comparisons use position and length rather than area or angle, since position and length are easier to compare accurately. Third, the dashboard avoids unnecessary 3D effects, decorative textures, and overloaded legends.

The dashboard also uses sorting strategically. Regional bars are sorted by percentage regulated, and lollipop charts are ranked by DALY or death rate. Sorting improves cognitive efficiency by allowing users to identify extremes without scanning unordered categories.

A design limitation is that the choropleth map may visually emphasize large countries more than small countries, even though each country is one observation. This is partially addressed by including KPI cards, count charts, regional summaries, and lollipop charts.

7. Tables, Spatial Data, Networks, and Maps

7.1 Spatial Data and Maps

Spatial data is central to the project because regulation status is country-level and geographically meaningful. The choropleth map enables detection of geographic clusters and gaps that would be difficult to identify from a table.

7.2 Tables

A raw table was not used as a primary visualization because the analytical goals focus on spatial and comparative reasoning rather than record-level lookup. However, the processed dataset remains available as a CSV output for reproducibility and inspection.

7.3 Networks and Trees

Networks and trees are not applicable to this project because the dataset does not contain relationship edges, hierarchical parent-child structures, or flow relationships. The units of analysis are countries, not linked nodes.

8. Validation

This project applies validation in four ways.

8.1 Data Validation

Data validation involved checking country names, ISO codes, missing values, and merge consistency. The preparation script includes a country-name crosswalk to reduce mismatches between IHME and regulation datasets, such as mapping "Russian Federation" to "Russia" and "Viét Nam" to "Vietnam." ISO-code validation was necessary because the choropleth map depends on ISO-3 country codes.

8.2 Task Validation

The visualizations were selected to match the stated analytical tasks. The map supports spatial pattern detection, the count chart supports summary comparison, the regional stacked bar supports proportional regional comparison, the scatterplot supports relationship exploration, the lollipop charts support ranking, and the heatmap supports multivariate regional summary.

8.3 Encoding Validation

The encoding choices were evaluated against perceptual effectiveness. Position and length were used for quantitative comparisons, color hue was used for categorical regulation status, and normalized color intensity was used in the heatmap for multivariate regional comparison.

8.4 Interpretation Validation

The dashboard includes a non-causal framing note: visual patterns show associations only and do not establish causal effects or measure enforcement quality. This is important because regulation presence does not necessarily imply strong enforcement, and health burden may be affected by many factors beyond just the lead paint regulation.

9. Analytical Insights and Knowledge Generation

The dashboard produces several meaningful insights.

9.1 Global Regulation Is Uneven

The KPI cards and country count chart show that 93 countries are regulated and 71 are not regulated in the processed dataset. This aligns with the broader global context (i.e. what WHO claimed as listed previously) that legally binding lead paint controls are still not universal.

9.2 Regional Disparities Are Clear

The regional stacked bar chart shows that Europe and South America have high shares of regulated countries, while Africa and Oceania have lower shares in the processed dashboard. This suggests that lead paint regulation is not evenly distributed across global regions.

9.3 Regulation Status and Health Burden Require Careful Interpretation

The lollipop charts show that some high-DALY countries are not regulated, while some high-death-rate countries are regulated. This is important because it demonstrates that legal regulation alone does not fully explain health burden. The dashboard therefore supports nuanced interpretation rather than a simplistic "regulated equals safe" message.

9.4 DALY Rate and Death Rate Are Positively Related

The scatterplot shows a positive relationship between lead-attributable DALY rate and death rate. This is expected because DALYs combine years of life lost and years lived with disability, while death rate captures mortality burden. OWID defines DALYs as a measure of total disease burden, where one DALY equals one lost year of healthy life.

9.5 Regional Heatmap Reveals Multivariate Patterns

The heatmap helps compare regulation share, average DALY rate, and average death rate simultaneously. Because indicators are normalized for color but displayed with actual values, the chart supports compact comparison without hiding the original values.

10. Limitations

Several limitations should be considered.

First, the project is observational and cross-sectional. It visualizes associations between regulation status and health-burden indicators but does not prove precisely that lead paint regulation causes lower DALY and/or death rates.

Second, regulation status is binary and does not capture enforcement strength, legal limits, policy scope, or compliance. OWID notes that the stringency of controls can vary by country, including differences in maximum allowed lead concentrations and product coverage.

Third, the health-burden indicators reflect lead exposure broadly, not only lead paint exposure (the project's main focus). Lead exposure can come from multiple sources such as paint, pipes, batteries, recycling, mining, smelting, contaminated dust, soil, water, and food.

11. Conclusion

This project demonstrates how data visualization can transform country-level policy and health-burden data into a coherent analytical dashboard. The final dashboard supports spatial exploration, categorical comparison, regional summarization, rates-relationship analysis, and ranked country-level burden detection.

The project shows that lead paint regulation remains uneven across countries and regions, and that high lead-attributable DALY and death-rate burdens persist in several countries. At the same time, the dashboard avoids overclaiming causality by clearly framing the results as visual associations rather than causal evidence.

Overall, the dashboard fulfills the course goals of problem formulation, data and task abstraction, justified encoding design, chart critique, spatial visualization, validation reasoning, and insight generation. The project also integrates IHME health-burden indicators with the OWID regulation dataset, moving beyond simple policy mapping toward a richer visual analysis of regulation and lead-related health burden.

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